

Non-parametric Statistical techniques

STA2020

Lecture 1

Introduction to Non-parametric techniques

Learning outcomes for this lecture

- 1) Understand what nonparametric statistical techniques are and when they are useful
 - I. Know the difference between parametric and nonparametric techniques
- 2) Understand the different data types and be able to classify any data into its type
- 3) Know how to rank a sample of data values
 - Understand how to deal with “ties” in the data values

Introduction to Non-Parametric Techniques

Parametric Techniques

- The statistical techniques introduced up to this point have all required that we know the underlying distribution of our sample (typically normal)
- or that the sample size is sufficient to rely on the normality conclusion of the central limit theorem.
- Based on these assumptions, the sampling distribution of test statistics were derived and we made inferences about unknown parameters.
- Also only considered quantitative data
- Examples: Estimating the population mean μ when the population variance σ^2 is unknown and sample size $n < 30$ OR hypothesis tests for σ^2 . Both ASSUME that the underlying population distribution is normally distributed

What about unknown distributions, small samples and qualitative data?

Non-Parametric techniques

- Make only weak assumptions about the underlying distribution of the data, e.g. might assume symmetry of the distribution
- Do not refer to any specific parameter of a population's distribution.
- Independence of observations and random samples are still required.
- Most of the test statistics do not depend on actual numerical values - they are performed on ranks of the data. Therefore, the actual specific distribution of the data does not matter and hence these tests are also called distribution-free statistics.

Non-Parametric tests

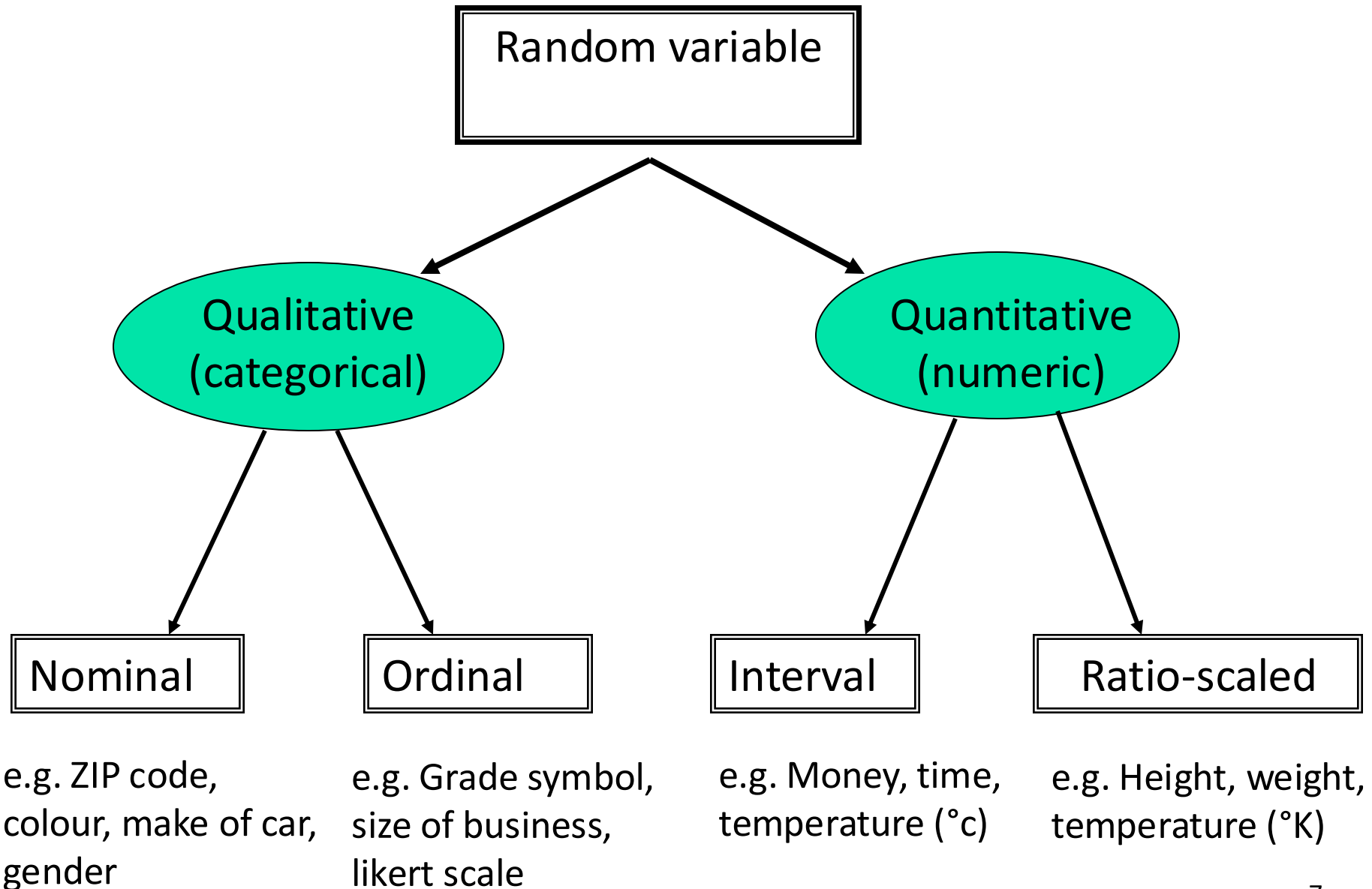
- Are used for non-normal (or when one has doubt about the precise distribution of) quantitative, nominal, and ordinal (ranked categories) data.

Note:

- If we know we have normally distributed quantitative data we may still use nonparametric techniques to analyze our data.
- However, in these cases we usually prefer to use parametric tests as they have more statistical power, since we know the appropriate model to use for the distribution.
- The power of a statistical test is the probability that the test will reject the null hypothesis when the alternative hypothesis is true (i.e. the probability of not committing a Type II error)

i.e. Nonparametric tests are always valid. Though in some circumstances they are not optimal in power.

Review : Data Types



Data Types

- Nominal data can be placed into categories, but these categories do not have a natural order.
- Ordinal data can be placed into categories and can be ranked with respect to some characteristic. However, we cannot interpret the difference between ranked values as numbers used are arbitrary.
- Interval-scale data has all the characteristics of ordinal data AND the differences between any two values have meaning. However, there is no “absolute” zero point on the scale and thus ratios between values are not meaningful.
- Ratio-scale data has all the characteristics of interval-scale data AND has a true zero point as its origin. Hence, ratios between values are meaningful.

Interval vs Ratio

- Is time an interval or ratio variable?
- Time has a natural order
- Differences between points in time are equal, so difference between 12pm and 1pm is the same as 4pm and 5pm (i.e. one hour).
- BUT time does not have a meaningful zero, there is not zero time point
- So can't say 2pm is twice as "old of a time" than 1pm – NOT MEANINGFUL!
- Ratio is not meaningful!
- What about duration or length of time, e.g. seconds?
- There is a natural order, so 1 second is smaller than 2 seconds
- Equal distances between points, so the difference between 10 and 15 seconds is the same as between 5 and 10 seconds.
- AND it has a true zero, 0 seconds is meaningful and thus, so are ratios (10 seconds is twice as long as 5 seconds)
- Go through the same reasoning for temperature in Celsius vs in Kelvin

Data Types --- Example

Determine the type of data for the following:

- 1) The number of students in a statistics class
- 2) The make of car driven by each of a sample of executives
- 3) The rating (Extremely poor[1], Very Poor[2], Poor[3], Unsure[4], Good[5], Very Good[6], Excellent[7]) reported for a particular television program by each of a sample of viewers
- 4) The weekly closing price of gold throughout the year
- 5) The month of highest sales for each firm in a sample

Data Types --- Example

Determine the type of data for the following:

- 6) The socioeconomic status of people who reside in Cape Town (upper class, middle class, lower class)
- 7) The responses by citizens on a 5-point rating scale (where 1=Strongly Disagree, 2=Disagree, 3=Unsure, 4=Agree, 5=Strongly Agree) to the statement: “South Africa should be divided into two time zones”
- 8) The gender of UCT employees.
- 9) The maximum temperature recorded in March 2013 (in °C)
- 10) The rating (excellent, good, fair or poor) given to a particular television program by each of a sample of viewers

Overview of Non-parametric Tests

Tests for a SINGLE population:

Test	Data type	Data
Tests for randomness of order (RUNS test)	Nominal	Independent observations
Chi-square Goodness-of-fit test	Nominal	Independent observations

Overview of Non-parametric Tests

Tests to compare TWO populations:

Tests for equality of <u>medians</u>	Data type	Data	Corresponding parametric test
Wilcoxon Rank Sum (also called Mann-Whitney)	Ordinal or non-normal quantitative	Independent samples	t-test for difference of means
Wilcoxon Signed Rank Sum	Non-normal quantitative	Matched/paired samples	matched pairs t-test for difference of means
Sign Test	Ordinal	Matched/paired/dependent samples	matched pairs t-test for difference of means

Overview of Non-parametric Tests

Tests to compare THREE OR MORE populations:

Test	Data type	Data	Corresponding parametric test
Kruskal-Wallis	Ordinal or non-normal quantitative	Independent samples	Single factor one way ANOVA
Friedman	Ordinal or non-normal quantitative	Matched/blocked dependent samples in experimental design	Randomized block ANOVA

Overview of Non-parametric Tests

Tests for the relationship between two variables:

Test	Data type	Data	Corresponding parametric test
Spearman's Rank Correlation	Ordinal or non-normal quantitative	Two random samples	Pearson's correlation coefficient

Ranking of Data

To rank data, we

- order/arrange the data in some sequence (usually in ascending order).
- identify the relative position of each value in the ordered data.
- if there are no ties in the data (a “tie” occurs when two or more observations have the same value), then rank = relative position.
- if there are ties in the data, then rank of the tied observations = average of relative positions of tied observations.

Ranking Data --- Example 1

Rank the following values:

. 4 9 6 7 5 2 8

Ranking Data --- Example 2

Rank the following values:

• 29 18 29 19 20 21 34 33 30 23

Ranking Data --- Example 3

Rank the following values:

29 18 29 19 20 21 20 33 30 23 33 33 24

Lecture 2

Wilcoxon Signed Rank Sum Test

Quick recap of lecture 1

1. **Parametric techniques** are based on the idea that your data come from a population that has a **known or hypothesized distribution**.
2. **Non-parametric techniques** are used for data that do not belong to a known or hypothesized distribution, they are distribution free, and instead **use ranks, signs** or frequencies of data.
3. **Data types:**
 1. Qualitative (Categorical): Nominal and Ordinal
 2. Quantitative (Numeric): Interval and Ratio
4. **Ranking of data:** order, rank and deal with ties

Learning outcomes for this lecture

- 1) Know the conditions (type of data, number of samples, relationship between the samples and objective) under which it is appropriate to conduct the WILCOXON SIGNED RANK SUM test
- 3) Know the assumptions and objective of the WILCOXON SIGNED RANK SUM test and be able to conduct the test for two matched/paired populations

Wilcoxon SIGNED Rank Sum Test

When is it appropriate? *When you want to compare two matched/paired samples of quantitative (interval or ratio-scaled) data with respect to their central locations (typically their medians) i.e. determine whether they come from the same population.*

- Remember: paired t-test we looked at the mean of differences between the samples
- For the Wilcoxon signed rank sum test (the equivalent non-parametric), we look at the **median** (central tendency) of differences

H_0 : The location of paired differences = 0 / The median difference is equal to zero = 0

- In other words, the samples come from the same population, there is no difference between the samples.

Wilcoxon SIGNED Rank Sum Test

Alternative hypothesis:

Either H_1 : The median/location of paired difference is $\neq 0$ (two-sided test)

OR H_1 : The median/location of paired difference > 0 (one-sided test)

OR H_1 : The median/location of paired difference < 0 (one-sided test)

Note that you are required to state these hypotheses using information from the question context i.e. do NOT simply state these theoretical hypotheses in a test or exam!

Wilcoxon SIGNED Rank Sum Test

Data and Assumptions:

- Two paired samples.
- Data is quantitative but not normal.
- Under H_0 , the distribution of the population of differences within pairs is symmetric around the median.
- The n paired differences are independent.

Wilcoxon SIGNED Rank Sum Test

Calculating the test statistic:

- 1) Calculate the difference for each pair
- 2) Eliminate all differences that are equal to 0
- 3) n = number of non-zero paired differences
- 4) Record the sign of the paired differences
- 5) Rank the absolute value of the paired differences
- 6) W = the sum of the signed ranks of the paired differences

Note: it is useful to construct a Table to complete steps 1 - 5

Logic behind the test

Null hypothesis = median difference is zero / no difference between two paired conditions/measurements

- So, under H_0 , we would expect distribution of differences between paired observations to be symmetric around zero (median of zero).
- i.e. as many positive differences as negative differences
- Example: Comparing weight of people before and after a weight loss drug. If the drug was not effective, we would expect to see roughly equal number of people who lost weight and gained weight.
- i.e. no consistent pattern in differences
- This test ranks the absolute values of these differences regardless of their sign first and then re-applies the sign afterwards.
- If H_0 is true, then the distribution of positive and negative ranks should be random across the ranks
- No systematic bias where larger or smaller ranks are consistently positives or negative
= sum of signed ranks (W) should be close to zero!

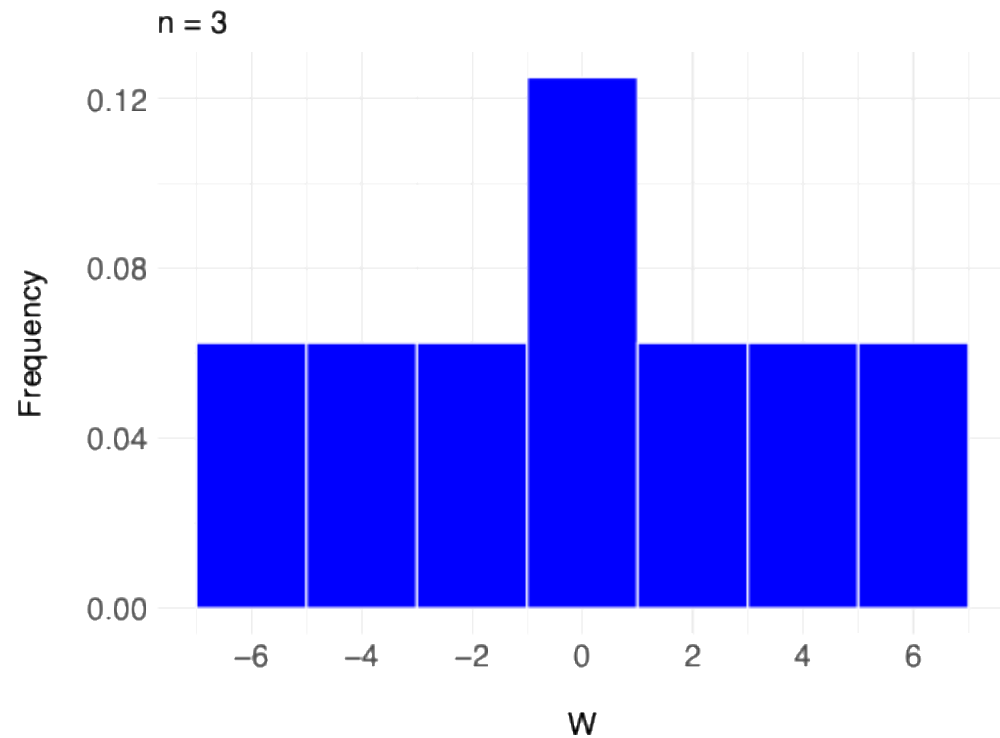
Logic behind the test

- If larger ranks are more positive, implies that first sample tends to be greater than the second sample and that null hypo is not valid.
- Expect to see W that is greater than zero
- Continuing with the weight loss example, after ranking the differences in weight (ignoring the sign), you'd expect the positive and negative differences to be mixed randomly among the ranks if the drug was ineffective.
- If you consistently saw that all large ranks (indicating large differences) were positive, this might suggest a systematic change (like the diet working), which would contradict the null hypothesis.
- Like before, we ask how far away from zero does W have to be (in either direction for two-sided) to reject the null hypothesis?

Sampling distribution of W

- For small n , it is possible to deduce the properties of the sampling distribution of W through simple enumeration of all possibilities

Ranks			
1	2	3	W
+	+	+	+6
-	+	+	+4
+	-	+	+3
+	+	-	0
-	-	+	0
-	+	-	-2
+	-	-	-4
-	-	-	-6



Sampling distribution of W

- So for small n , we can get exact critical values and so there is a specific table of critical values to conduct hypothesis test
- BUT it turns out, that for $n > 10$, the sampling distribution of W starts to become normal!
- $Z = \frac{W - \mu_T}{\sigma_T}$ where $\mu_T = 0$ and $\sigma_T = \sqrt{\frac{n(n+1)(2n+1)}{6}}$

Critical Region:

Reject H_0 if $|z| \geq z_{\alpha/2}$ (two-sided test) OR

if $z > z_{\alpha}$ (one-sided $>$ test) OR

if $z < -z_{\alpha}$ (one-sided $<$ test)

where z_{α} is the critical value
(critical value approach)

OR if the calculated p-value $\leq \alpha$
(p-value approach)

Example 1

Does a “flexi-time” work-schedule help reduce the travel time of workers to work?

A random sample of 32 workers was selected, and workers recorded their travel time in minutes before and after the program was implemented.

(Test at the 5% significance level, using the modified (p-value) approach)

Wilcoxon Signed Rank Sum Test --- Example 1

Worker	8:00-Arr	Flextime
1	34	31
2	35	31
3	43	44
4	46	44
5	16	15
6	26	28
7	68	63
8	38	39
9	61	63
10	52	54
11	68	65
12	13	12
13	69	71
14	18	13
15	53	55
16	18	19

Worker	8:00-Arr	Flextime
17	41	38
18	25	23
19	17	14
20	26	21
21	44	40
22	30	33
23	19	18
24	48	51
25	29	33
26	24	21
27	51	50
28	40	38
29	26	22
30	20	19
31	19	21
32	42	38

Wilcoxon Signed Rank Sum Test --- Example 1

We want to compare two matched/paired samples of quantitative data. The samples do not adhere to a normal distribution, and we are interested in whether there is a difference between them, so we use **the Wilcoxon Signed Rank Sum test**.

H_0 : There is no difference in travel time to work between the normal work-hour and flexi-time work programs. The median difference is zero.

H_1 : Workers take longer to travel to work in the normal work-hour program i.e. workers take less time to travel to work in the flexi-time program (hence flexitime does reduce workers' travel time). The median difference is greater than zero.

Example 1

Worker	8:00-Arr	Flextime	Diff	Diff
1	34	31	3	3
2	35	31		
3	43	44		
4	46	44		
5	16	15		
6	26	28		
7	68	63		
8	38	39		
9	61	63		
10	52	54		
11	68	65		
12	13	12		
13	69	71		
14	18	13		
15	53	55		
16	18	19		

Worker	8:00-Arr	Flextime	Diff	Diff
17	41	38		
18	25	23		
19	17	14		
20	26	21		
21	44	40		
22	30	33		
23	19	18		
24	48	51		
25	29	33		
26	24	21		
27	51	50		
28	40	38		
29	26	22		
30	20	19		
31	19	21		
32	42	38	4	4

Example 1

Worker	8:00-Arr	Flextime	Diff	Diff
1	34	31	3	3
2	35	31	4	4
3	43	44	-1	1
4	46	44	2	2
5	16	15	1	1
6	26	28	-2	2
7	68	63	5	5
8	38	39	-1	1
9	61	63	-2	2
10	52	54	-2	2
11	68	65	3	3
12	13	12	1	1
13	69	71	-2	2
14	18	13	5	5
15	53	55	-2	2
16	18	19	-1	1

Worker	8:00-Arr	Flextime	Diff	Diff
17	41	38	3	3
18	25	23	2	2
19	17	14	3	3
20	26	21	5	5
21	44	40	4	4
22	30	33	-3	3
23	19	18	1	1
24	48	51	-3	3
25	29	33	-4	4
26	24	21	3	3
27	51	50	1	1
28	40	38	2	2
29	26	22	4	4
30	20	19	1	1
31	19	21	-2	2
32	42	38	4	4

Example 1

Worker	Diff	Diff	Rank
1	3	3	
2	4	4	
3	-1	1	
4	2	2	
5	1	1	
6	-2	2	
7	5	5	
8	-1	1	
9	-2	2	
10	-2	2	
11	3	3	
12	1	1	
13	-2	2	
14	5	5	
15	-2	2	
16	-1	1	

Worker	Diff	Diff	Rank
17	3	3	
18	2	2	
19	3	3	
20	5	5	
21	4	4	
22	-3	3	
23	1	1	
24	-3	3	
25	-4	4	
26	3	3	
27	1	1	
28	2	2	
29	4	4	
30	1	1	
31	-2	2	
32	4	4	

Example 1

Worker	Diff	Diff	Rank
1	3	3	
2	4	4	
3	-1	1	
4	2	2	
5	1	1	
6	-2	2	
7	5	5	
8	-1	1	
9	-2	2	
10	-2	2	
11	3	3	
12	1	1	
13	-2	2	
14	5	5	
15	-2	2	
16	-1	1	

Worker	Diff	Diff	Rank
17	3	3	
18	2	2	
19	3	3	
20	5	5	
21	4	4	
22	-3	3	
23	1	1	
24	-3	3	
25	-4	4	
26	3	3	
27	1	1	
28	2	2	
29	4	4	
30	1	1	
31	-2	2	
32	4	4	

8-way tie assign the average of ranks 1 to 8 to each value:

Average =

$$\frac{1+2+3+4+5+6+7+8}{8}$$



Example 1

Worker	Diff	Diff	Rank
1	3	3	21
2	4	4	27
3	-1	1	4.5
4	2	2	13
5	1	1	4.5
6	-2	2	13
7	5	5	31
8	-1	1	4.5
9	-2	2	13
10	-2	2	13
11	3	3	21
12	1	1	4.5
13	-2	2	13
14	5	5	31
15	-2	2	13
16	-1	1	4.5

Worker	Diff	Diff	Rank
17	3	3	21
18	2	2	13
19	3	3	21
20	5	5	31
21	4	4	27
22	-3	3	21
23	1	1	4.5
24	-3	3	21
25	-4	4	27
26	3	3	21
27	1	1	4.5
28	2	2	13
29	4	4	27
30	1	1	4.5
31	-2	2	13
32	4	4	27

Example 1

Worker	Diff	Diff	Rank	Signed Rank
1	3	3	21	+21
2	4	4	27	+27
3	-1	1	4.5	-4.5
4	2	2	13	+13
5	1	1	4.5	+4.5
6	-2	2	13	-13
7	5	5	31	+31
8	-1	1	4.5	-4.5
9	-2	2	13	-13
10	-2	2	13	-13
11	3	3	21	+21
12	1	1	4.5	+4.5
13	-2	2	13	-13
14	5	5	31	+31
15	-2	2	13	-13
16	-1	1	4.5	-4.5

Worker	Diff	Diff	Rank	Signed Rank
17	3	3	21	+21
18	2	2	13	+13
19	3	3	21	+21
20	5	5	31	+31
21	4	4	27	+27
22	-3	3	21	-21
23	1	1	4.5	+4.5
24	-3	3	21	-21
25	-4	4	27	-27
26	3	3	21	+21
27	1	1	4.5	+4.5
28	2	2	13	+13
29	4	4	27	+27
30	1	1	4.5	+4.5
31	-2	2	13	-13
32	4	4	27	+27

Test statistic

n = number of non-zero differences = $32 > 10$

Test Statistic: W = sum of ranked signed paired differences = 207

$$\Rightarrow z = \frac{W - \mu_W}{\sigma_W} \text{ where } \mu_W = 0 \text{ and } \sigma_W = \sqrt{\frac{n(n+1)(2n+1)}{6}} = \sqrt{\frac{32(33)(65)}{6}} = 106.958$$

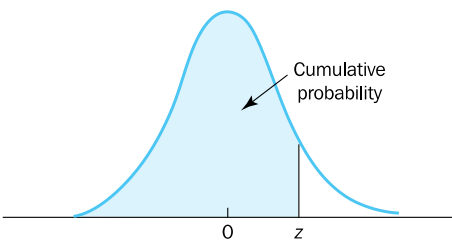
$$\Rightarrow z = \frac{207 - 0}{106.958} = 1.935$$

Critical Region: Reject H_0 if $p\text{-value} \leq 0.05$

p -value of test statistic?

Use standard normal distribution table...

TABLE 1 (Continued)



Entries in the table give the area under the curve to the left of the z value. For example, for z = 1.25, the cumulative probability is .8944.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9913
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9986	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990

Test statistic = 1.935 ≈ 1.94

p-value = 1 – 0.9738 = 0.0262

Conclusion

Conclusion: Since $p\text{-value} = 0.0262 < 0.05$ we Reject H_0 at the 5% significance level and conclude that workers do take longer to travel to work during normal hours than flexitime i.e. there is sufficient evidence that flexi-time does reduce the travel time of workers to work.

Example 2

Context

Suppose that 16 students are asked two basic probability questions (question A and B) and to give their immediate and intuitive answer without doing any calculations. The instructor (person asking the questions) hypothesizes that if students have developed a good understanding of probability theory, then they will give higher probabilities for question A than B. If they have not listened in class, then their answers will be random and there will be no overall tendency either way.

H0: The median of the differences between the probabilities given for question A and question B is zero. There is no difference between the probabilities given for question A and B.

H1: The median of the differences between the probabilities given for question A and question B is greater than zero. The probabilities given for question A are higher than that of question B.

Results

Subj.	X _A	X _B	X _A —X _B
1	78	78	0
2	24	24	0
3	64	62	+2
4	45	48	-3
5	64	68	-4
6	52	56	-4
7	30	25	+5
8	50	44	+6
9	64	56	+8
10	50	40	+10
11	78	68	+10
12	22	36	-14
13	84	68	+16
14	40	20	+20
15	90	58	+32
16	72	32	+40
median difference = +5.5			

1	2	3	4	5	6	7
Subject	X_A	X_B	$X_A - X_B$	$ X_A - X_B $	Abs rank	Signed rank
1	78	78	0	0	---	---
2	24	24	0	0	---	---
3	64	62	+2	2	1	+1
4	45	48	-3	3	2	-2
5	64	68	-4	4	3.5	-3.5
6	52	56	-4	4	3.5	-3.5
7	30	25	+5	5	5	+5
8	50	44	+6	6	6	+6
9	64	56	+8	8	7	+7
10	50	40	+10	10	8.5	+8.5
11	78	68	+10	10	8.5	+8.5
12	22	36	-14	14	10	-10
13	84	68	+16	16	11	+11
14	40	20	+20	20	12	+12
15	90	58	+32	32	13	+13
16	72	32	+40	40	14	+14
W = 67.0, n = 14						

Conclusion

n = number of non-zero differences = $14 > 10$

Test Statistic: W = sum of ranked signed paired differences = 67

$$\Rightarrow z = \frac{W - \mu_W}{\sigma_W} \text{ where } \mu_W = 0 \text{ and } \sigma_W = \sqrt{\frac{n(n+1)(2n+1)}{6}} = \sqrt{\frac{14(15)(29)}{6}} = 31.86$$

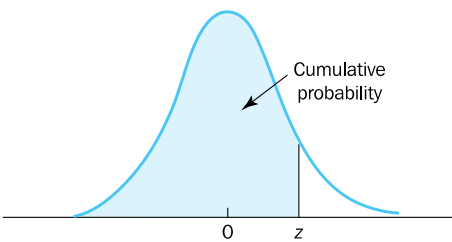
$$\Rightarrow z = \frac{67 - 0}{31.86} = 2.10$$

Critical Region: Reject H_0 if p-value ≤ 0.05

p-value of test statistic?

Use standard normal distribution table...

TABLE 1 (Continued)



Entries in the table give the area under the curve to the left of the z value. For example, for z = 1.25, the cumulative probability is .8944.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9913
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9986	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990

Test statistic = 2.1

p-value = 1 – 0.9821 = 0.0179

Conclusion

Since $p\text{-value} = 0.0179 < 0.05$ we Reject H_0 at the 5% significance level and conclude that the students : The median of the differences between the probabilities given for question A and question B is greater than zero.

The probabilities given for question A are higher than that of question B / students did give higher probabilities for question A.

Lecture 3

Mann-Whitney Wilcoxon Test

Quick recap of lecture 2

1. **Wilcoxon Signed Rank Sum test**, is used for two paired samples of non-normal quantitative data
2. Tests the null hypothesis that the median difference is zero
3. It achieves this **by ranking the differences** between samples, re-applying **sign of the difference** to the ranks and calculating the **sum of signed ranks** as the test statistic.
4. For **sample sizes greater than 10**, this **test statistic is approximately normally distributed** so can use a z-score to draw conclusions about null hypothesis.

Learning outcomes for this lecture

- 1) Know the differences between the Wilcoxon Signed Rank Sum Test and the Mann-Whitney-Wilcoxon Test
- 2) Know the conditions (data type, number of samples, relationship between the samples and objective) under which it is appropriate to conduct the Mann-Whitney-Wilcoxon test
- 3) Know the data assumptions of the Mann-Whitney-Wilcoxon Test
- 4) Be able to conduct the Mann-Whitney-Wilcoxon Test

Mann-Whitney-Wilcoxon Test

- Also called the U-test, Wilcoxon Rank Sum Test, or just Rank Sum Test.
- **When is it appropriate?** The objective of this test is to determine whether two *independent* samples of ordinal or quantitative data have the same location (median) i.e. are from the same population
 - It is the equivalent of the t-test for two samples of normal data.
- **Data and Assumptions:**
 - 2 random samples of size n_1 and n_2 .
 - The data are either ordinal or quantitative (but not normal)
 - Samples and observations within samples are independent.
 - The distributions of the two populations differ with respect to location only (if they differ at all).

Mann-Whitney-Wilcoxon Test

Hypotheses:

H_0 : The two population locations are the same / identical distributions

Either H_1 : The two populations differ in locations
(two-sided test)

OR H_1 : The location of the first population is to the right of
the location of the second population ($>$ --- one-sided test)

OR H_1 : The location of the first population is to the left of
the location of the second population ($<$ --- one-sided test)

Mann-Whitney-Wilcoxon Test

Calculating the Test statistic:

The test statistic depends on n_1 (number of observations in the first sample) and n_2 (number of observations in the second sample)

- 1) Combine the two samples
- 2) Rank all observations from smallest to largest, that is, from 1 to n_1+n_2
- 3) T_1 = sum of ranks of the first sample and T_2 = sum of ranks of the second sample

Note: $T_1 + T_2 = \frac{(n_1 + n_2)(n_1 + n_2 + 1)}{2}$

Mann-Whitney-Wilcoxon Test

- If the locations of the two populations are about the same (i.e. the null hypothesis is true) we would expect the ranks to be evenly spread between the samples and the sum of the ranks to be similar.
- If T_1 is sufficiently small, then most of the smaller observations are in population 1
 - The location of population 1 is to the left of the location of population 2
 - Reject the null hypothesis
- If T_1 is sufficiently large, then most of the larger observations are in population 1
 - the location of population 1 is to the right of the location of population 2
 - Reject the null hypothesis.

Example

- **Sample 1** = 011012123
- **Sample 2** = 7 9 10 8 10 11 10 11 12

0	1	1	0	1	2	1	2	3	7	9	10	8	10	11	10	11	12
1.5	4.5	4.5	1.5	4.5	7.5	4.5	7.5	9	10	12	14	11	14	16. 5	14	16. 5	18

- Clearly ranks are not evenly spread
- Sum of ranks for sample 1 = 45
- Sum of ranks for sample 2 = 126

Mann-Whitney-Wilcoxon Test

Small sample Wilcoxon Rank Sum Test (n_1 AND $n_2 < 10$)

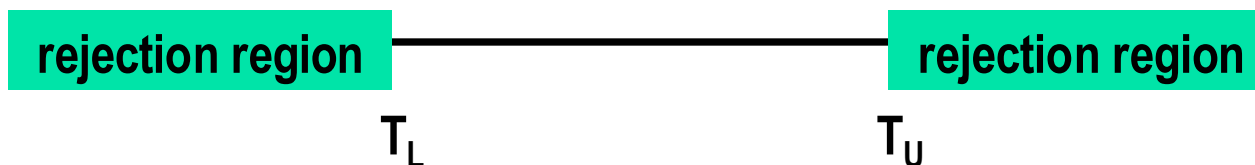
Test statistic: $T = T_1$

Critical Region:

- when n_1 AND $n_2 < 10$, critical values of T_L can be found in the appropriate table. The value of T_U is calculated as:

$$T_U = n_1(n_1 + n_2 + 1) - T_L$$

- Reject H_0 if $T \leq T_L$ or $T \geq T_U$



Mann-Whitney-Wilcoxon Test

Large sample Wilcoxon Rank Sum Test (n_1 AND/OR $n_2 \geq 10$)

Test Statistic: T is approximately normally distributed

$$Z = \frac{T - \mu_T}{\sigma_T} \quad \text{where} \quad \mu_T = \frac{n_1(n_1 + n_2 + 1)}{2} \quad \text{and} \quad \sigma_T = \sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}$$

Critical region:

Reject H_0 if $|z| \geq z_{\alpha/2}$ (two-sided test) OR

if $z > z_{\alpha}$ (one-sided > test) OR

if $z < -z_{\alpha}$ (one-sided < test)

where z_{α} is the critical value
(critical value approach)

OR if the calculated p-value $\leq \alpha$
(p-value approach)

Mann-Whitney-Wilcoxon Test

The ABC Company has sent 13 of its employees to a privately-run programme providing word-processing skills training. Six of the employees were from the data-processing (DP) department and the others were from the typing (T) pool.

At the end of the programme, the company received a report indicating the score received by each of its employees out of a total possible score of 100. The scores of the 13 employees of ABC are given in the following table (next slide):

Is there a difference in the performance of the two groups in the word-processing programme? Test at a 5% significance level.

Mann-Whitney-Wilcoxon Test --- Example 1

DP	T
70	59
52	70
46	75
65	85
60	50
40	82
	64

Mann-Whitney-Wilcoxon Test --- Example 1

We want to compare two **independent** samples of quantitative data with respect to their locations use the Mann-Whitney-Wilcoxon test.

H_0 : There is no difference in performance between the two groups on the word processing program

H_1 : There is a difference in performance between the two groups on the word processing program

$$n_1 = 6$$

$$n_2 = 7$$

n_1 AND $n_2 < 10$ ☐ Small Sample Test

Mann-Whitney-Wilcoxon Test--- Example 1

Calculating the Test Statistic:

DP	T	Combined Sample	Ordered Combined Sample	Rank
70	59	70		
52	70	52		
46	75	46		
65	85	65		
60	50	60		
40	82	40		
	64	59		
		70		
		75		
		85		
		50		
		82		
		64		

Mann-Whitney-Wilcoxon Test --- Example 1

Calculating the Test Statistic:

DP	T	Combined Sample	Ordered Combined Sample	Rank
70	59	70	40	1
52	70	52	46	2
46	75	46	50	3
65	85	65	52	4
60	50	60	59	5
40	82	40	60	6
	64	59	64	7
		70	65	8
		75	70	9.5
		85	70	9.5
		50	75	11
		82	82	12
		64	85	13

$$T_1 = 1 + 2 + 4 + 6 + 8 + 9.5 = 30.5$$

$$T_2 = 3 + 5 + 7 + 9.5 + 11 + 12 + 13 = 60.5$$

Critical Region: Reject H_0 if $T \leq T_L$ or $T \geq T_U$

TABLE 8 T_L Values for the Mann–Whitney–Wilcoxon test

Reject the hypothesis of identical populations if the sum of the ranks for the n_1 items is *less* than the value T_L shown in the following table or if the sum of the ranks for the n_1 items is *greater* than the value T_U where:

$$T_U = n_1(n_1 + n_2 + 1) - T_L$$

$\alpha = 0.05$	n_2								
n_1	2	3	4	5	6	7	8	9	10
2	3	3	3	3	3	3	4	4	4
3	6	6	6	7	8	8	9	9	10
4	10	10	11	12	13	14	15	15	16
5	15	16	17	18	19	21	22	23	24
6	21	23	24	25	27	28	30	32	33
7	28	30	32	34	35	37	39	41	43
8	37	39	41	43	45	47	50	52	54
9	46	48	50	53	56	58	61	63	66
10	56	59	61	64	67	70	73	76	79

$$n_1 = 6$$

$$n_2 = 7$$

$$T_L = 28$$

$$T_U = n_1(n_1 + n_2 + 1) - T_L = 6(6 + 7 + 1) - 28 = 56$$

Mann-Whitney-Wilcoxon Test --- Example 1

Critical Region: Reject H_0 if $T \leq 28$ or $T \geq 56$

Conclusion: Since $28 < T < 56$ we do NOT Reject H_0 and we conclude that there is no significant difference in the performance of the two groups in the word-processing programme at the 5% significance level

Mann-Whitney-Wilcoxon Test --- Example 2

A pharmaceutical company is planning to introduce a new painkiller. To determine the effectiveness of the drug in comparison to aspirin, 30 people were randomly selected.

- 15 were given the new drug (Sample 1).
- 15 were given aspirin (Sample 2).

Each participant was asked to indicate which one of five statements best represented the effectiveness of the drug they took.

The drug taken was...
(5) Extremely effective
(4) Quite effective
(3) Somewhat effective
(2) Slightly effective
(1) Not at all effective

Mann-Whitney-Wilcoxon Test --- Example 2

New	Aspirin
3	4
5	1
4	3
3	2
2	4
5	1
1	3
4	4
5	2
3	2
3	2
5	4
5	3
5	4
4	5

Summary of Results:

The drug's score was...	DRUG	
	New	Aspirin
5	6	1
4	3	5
3	4	3
2	1	4
1	1	2

Can we conclude that the new painkiller is perceived to be more effective?

Test at the 5% significance level

Mann-Whitney-Wilcoxon Test --- Example 2

We want to compare two independent samples of ordinal data with respect to their locations use the Mann-Whitney-Wilcoxon test.

H_0 : There is no difference in effectiveness between the two painkillers.

H_1 : The new painkiller is more effective (one-sided $>$ test).

$$n_1 = 15$$

$$n_2 = 15$$

n_1 and $n_2 > 10$ Large Sample Test

Mann-Whitney-Wilcoxon Test --- Example 2

Calculating the Test Statistic:

Combined Sample			
New	Rank	Aspirin	Rank
3	12	4	19.5
5	27	1	2
4	19.5	3	12
3	12	2	6
2	6	4	19.5
5	27	1	2
1	2	3	12
4	19.5	4	19.5
5	27	2	6
3	12	2	6
3	12	2	6
5	27	4	19.5
5	27	3	12
5	27	4	19.5
4	19.5	5	27

Mann-Whitney-Wilcoxon Test --- Example 2

Combined Sample			
New	Rank	Aspirin	Rank
3	12	4	19.5
5	27	1	2
4	19.5	3	12
3	12	2	6
2	6	4	19.5
5	27	1	2
1	2	3	12
4	19.5	4	19.5
5	27	2	6
3	12	2	6
3	12	2	6
5	27	4	19.5
5	27	3	12
5	27	4	19.5
4	19.5	5	27
T_1	276.5	T_2	188.5

$$\text{Check: } T_1 + T_2 = 465 \quad \frac{(n_1 + n_2)(n_1 + n_2 + 1)}{2} = \frac{(15 + 15)(15 + 15 + 1)}{2} = 465$$

Mann-Whitney-Wilcoxon Test --- Example 2

Test Statistic: $T = T_1 = 276.5$

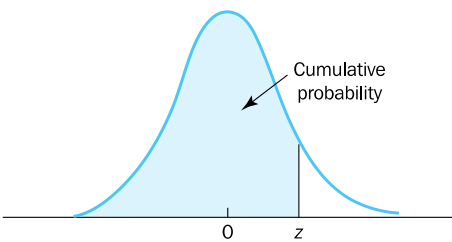
$$\Rightarrow Z = \frac{T - \mu_T}{\sigma_T} \quad \text{where: } \mu_T = \frac{n_1(n_1 + n_2 + 1)}{2} = \frac{15(15 + 15 + 1)}{2} = 232.5$$

$$\sigma_T = \sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}} = \sqrt{\frac{(15)(15)(15 + 15 + 1)}{12}} = 24.11$$

$$\Rightarrow Z = \frac{276.5 - 232.5}{24.11} = 1.82$$

Critical Region: Reject H_0 if $z \geq 1.645$ OR approximate p-value ≤ 0.05

TABLE 1 (Continued)



Entries in the table give the area under the curve to the left of the z value. For example, for z = 1.25, the cumulative probability is .8944.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9913
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9986	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990

Test statistic = 1.82

p-value = 1 – 0.9656 = 0.0344

Mann-Whitney-Wilcoxon Test --- Example 2

p-value ≈ 0.0344

Conclusion (a): Since $1.82 > 1.645$ we Reject H_0 and conclude that the new painkiller is more effective at the 5% level of significance

OR

Conclusion (b): Since p-value $\approx 0.0344 < 0.05$ we Reject H_0 and conclude that the new painkiller is more effective at the 5% level of significance

Lecture 4

Kruskal-Wallis test

Quick recap of lecture 3

1. **Mann Whitney U test** is used for two independent samples of either ordinal or non-normal quantitative data.
2. Tests the null hypothesis that **the medians of the two populations are the same**.
3. It achieves this by combining the samples, ranking their values and calculating the sum of ranks for each sample.
4. For **small sample** sizes (< 10), use **table of exact probabilities**.
5. For **larger sample** sizes, the **test statistic** is **approximately normally distributed** and can use z-score.

Learning outcomes for this lecture

- 1) Know the conditions (data type, number of samples, relationship between the samples and objective) under which it is appropriate to conduct the Kruskal-Wallis test
- 2) Know the data assumptions of the Kruskal-Wallis test
- 3) Be able to conduct the Kruskal-Wallis test

Kruskal-Wallis Test

When is it appropriate? When you want to compare more than two independent groups/samples of ordinal or quantitative data with respect to their locations (medians) i.e. determine whether they all come from the same population.

- The test is an equivalent of a single factor ANOVA.

Data and Assumptions:

- The data are either ordinal or quantitative but not necessarily normal.
- The treatment levels and observations within treatment levels are independent.
- There are at least three observations per sample ($n_i \geq 3$).
- The distributions of all the populations differ with respect to location only (if they differ at all).

Kruskal-Wallis Test

Hypotheses:

H_0 : The locations of all the k populations (groups) are the same.

H_1 : At least two population locations differ.

Calculating the Test statistic:

- 1) Combine observations from all k groups to form one sample ($n_T = \sum n_j$)
- 2) Rank the observations from 1 (smallest) to n_T (largest).
- 3) Average ranks of tied observations.
- 4) Calculate the rank sums $T_1, T_2, \dots, T_k$ for all the k groups (treatments).

$$(\text{check that } \sum_{i=1}^k T_i = \frac{n_T(n_T+1)}{2})$$

Kruskal-Wallis Test

Test statistic:
$$H = \left[\frac{12}{n_T(n_T+1)} \sum_{i=1}^k \frac{T_i^2}{n_i} \right] - 3(n_T + 1)$$

Note: If all the populations have the same location (H_0 is true), the ranks should be evenly distributed among the k samples and the statistic H will be small. For example:

Uneven distribution of ranks

Sample 1	Sample 2	Sample3
1 (1)	4 (4)	7 (7)
2 (2)	5 (5)	8 (8)
3 (3)	6 (6)	9 (9)
$T_1=6$	$T_2=15$	$T_3=24$

$$H = 7.2$$

Even distribution of ranks

Sample 1	Sample 2	Sample3
1 (1)	2 (2)	3 (3)
4 (4)	5 (5)	6 (6)
9 (9)	8 (8)	7 (7)
$T_1=14$	$T_2=15$	$T_3=16$

$$H = 0.0888$$

Kruskal-Wallis Test

Critical Region:

- When the sample sizes (of all k groups) ≥ 3 , H is approximately chi-squared distributed with $k-1$ degrees of freedom.
- The test is one-sided in the sense that all of the critical region probability (i.e. the significance level) falls in the upper tail
- We reject the null hypothesis if H is too large. That is, reject H_0 if $H \geq c$, where c is the critical value, or approximate/exact p-value $\leq \alpha$.

i.e. Reject H_0 if: $\mathbf{H \geq \chi^2_{\alpha, k-1}}$ or approximate/exact p-value $\leq \alpha$.

Kruskal-Wallis Test --- Example 1

How do customers rate three shifts with respect to the speed of service in a particular 24-hr restaurant? Three samples of 10 customer response-cards were randomly selected, one sample from each shift, and customer ratings

(from 1 for very slow to 5 for very quick) were recorded:

Can we conclude that customers perceive the speed of service to be different among the three shifts at a 5% significance level?

4:00-mid	Mid-8:00	8:00-4:00
4	3	3
4	4	1
3	2	3
4	2	2
3	3	1
3	4	3
3	3	4
3	3	2
2	2	4
3	3	1

Kruskal-Wallis Test --- Example 1

We are wanting to compare three independent samples of ordinal data with respect to their locations (medians)

⇒ use the Kruskal-Wallis Test

H_0 : There is no difference in the perception customers have of the speed of service among the three shifts

H_1 : There is a difference in the perception customers have of the speed of service among the three shifts

Calculating the Test Statistic:

$$k = 3$$

$$n_1 = n_2 = n_3 = 10$$

Kruskal-Wallis Test --- Example 1

Ranking the COMBINED sample:

4:00 - mid	Mid - 8:00	8:00-4:00
4	3	3
4	4	1 (2)
3	2	3
4	2	2
3	3	1 (2)
3	4	3
3	3	4
3	3	2
2	2	4
3	3	1 (2)
T_1	T_2	T_3

Kruskal-Wallis Test --- Example 1

Ranking the COMBINED sample:

4:00 - mid	Mid - 8:00	8:00-4:00
4	3	3
4	4	1 (2)
3	2 (6.5)	3
4	2 (6.5)	2 (6.5)
3	3	1 (2)
3	4	3
3	3	4
3	3	2 (6.5)
2 (6.5)	2 (6.5)	4
3	3	1 (2)
T_1	T_2	T_3

Kruskal-Wallis Test --- Example 1

Ranking the COMBINED sample:

4:00 - mid	Mid - 8:00	8:00-4:00
4 (27)	3 (16.5)	3 (16.5)
4 (27)	4 (27)	1 (2)
3 (16.5)	2 (6.5)	3 (16.5)
4 (27)	2 (6.5)	2 (6.5)
3 (16.5)	3 (16.5)	1 (2)
3 (16.5)	4 (27)	3 (16.5)
3 (16.5)	3 (16.5)	4 (27)
3 (16.5)	3 (16.5)	2 (6.5)
2 (6.5)	2 (6.5)	4 (27)
3 (16.5)	3 (16.5)	1 (2)
186.5	156	122.5
T_1	T_2	T_3

Check: $\sum_{i=1}^k T_i = 465$

$$\frac{n(n+1)}{2} = \frac{(30)(31)}{2} = 465$$

Kruskal-Wallis Test --- Example 1

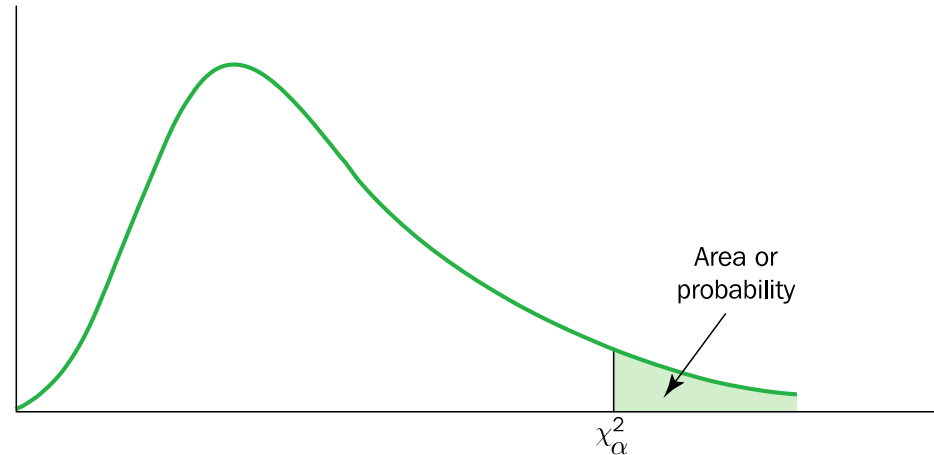
Test Statistic:

$$\begin{aligned} H &= \left[\frac{12}{n_T(n_T + 1)} \sum_{i=1}^k \frac{T_i^2}{n_i} \right] - 3(n_T + 1) \\ &= \left[\frac{12}{(30)(31)} \left(\frac{(186.5)^2}{10} + \frac{(156)^2}{10} + \frac{(122.5)^2}{10} \right) \right] - 3(31) \\ &= \frac{2}{155} (7412.45) - 93 \\ &= 2.645 \end{aligned}$$

Critical Region:

$$H \sim \chi^2_2$$

Reject H_0 if $H \geq \chi^2_{2:0.05}$ OR if the approximate p-value ≤ 0.05

TABLE 3 Chi-squared distribution

Entries in the table give χ^2_{α} values, where α is the area or probability in the upper tail of the chi-squared distribution. For example, with ten degrees of freedom and a .01 area in the upper tail, $\chi^2_{0.01} = 23.209$

Degrees of freedom	Area in upper tail									
	.995	.99	.975	.95	.90	.10	.05	.025	.01	.005
1	.000	.000	.001	.004	.016	2.706	3.841	5.024	6.635	7.879
2	.010	.020	.051	.103	.211	4.605	5.991	7.378	9.210	10.597
3	.072	.115	.216	.352	.584	6.251	7.815	9.348	11.345	12.838
4	.207	.297	.484	.711	1.064	7.779	9.488	11.143	13.277	14.860
5	.412	.554	.831	1.145	1.610	9.236	11.070	12.832	15.086	16.750
6	.676	.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.647	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188

Critical value = 5.991

p-value > 0.10

Kruskal-Wallis Test --- Example 1

Conclusion (a): Since $2.645 < 5.991$ we do NOT Reject H_0 and conclude that there is no significant difference in the customers' perceptions of the speed of service among the three shifts, at the 5% significance level

OR

Conclusion (b): Since $p\text{-value} > 0.10$ (> 0.05) we do NOT Reject H_0 and conclude that there is no significant difference in the customers' perceptions of the speed of service among the three shifts, at the 5% significance level

Kruskal-Wallis Test --- Example 2

To determine whether absentee rates are the same amongst three different levels of employees in a company, samples comprising of 4 top managers, 5 middle managers and 7 workers were selected and their records examined to determine how many days they reported sick in the last year.

Top	Middle	Worker
4	6	8
0	0	6
2	1	1
2	3	6
	2	5
		5
		6

Is there evidence that the absentee rates differ from one level of employee to another? Test at the 1% significance level

Kruskal-Wallis Test --- Example 2

We are wanting to compare three independent samples of quantitative data with respect to their locations (medians)

⇒ use the Kruskal-Wallis Test

H_0 : There is no difference in absenteeism rates from one level of employee to another

H_1 : There is a difference in absenteeism rates from one level of employee to another

Calculating the Test Statistic:

$$k = 3$$

$$n_1 = 4$$

$$n_2 = 5$$

$$n_3 = 7$$

Kruskal-Wallis Test --- Example 2

Ranking the COMBINED sample:

Top	Middle	Worker
4	6	8
0	0	6
2	1	1
2	3	6
	2	5
		5
		6
T_1	T_2	T_3

Kruskal-Wallis Test --- Example 2

Ranking the COMBINED sample:

Top	Middle	Worker
4 (9)	6 (13.5)	8 (16)
0 (1.5)	0 (1.5)	6 (13.5)
2 (6)	1 (3.5)	1 (3.5)
2 (6)	3 (8)	6 (13.5)
	2 (6)	5 (10.5)
		5 (10.5)
		6 (13.5)
22.5	32.5	81
T_1	T_2	T_3

$$\text{Check: } \sum_{i=1}^k T_i = 136 \qquad \frac{n(n+1)}{2} = \frac{(16)(17)}{2} = 136$$

Kruskal-Wallis Test --- Example 2

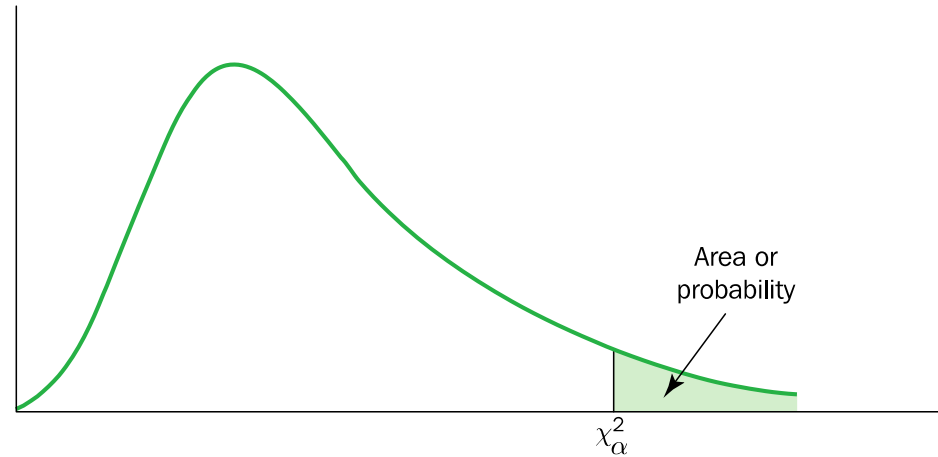
Test Statistic:

$$\begin{aligned} H &= \left[\frac{12}{n_T(n_T + 1)} \sum_{i=1}^k \frac{T_i^2}{n_i} \right] - 3(n_T + 1) \\ &= \left[\frac{12}{(16)(17)} \left(\frac{(22.5)^2}{4} + \frac{(32.5)^2}{5} + \frac{(81)^2}{7} \right) \right] - 3(17) \\ &= \frac{3}{68} (1275.098) - 51 \\ &= 5.254 \end{aligned}$$

Critical Region:

$$H \sim \chi^2_2$$

Reject H_0 if $H \geq \chi^2_{2;0.01}$ OR if the approximate p-value ≤ 0.01

TABLE 3 Chi-squared distribution

Entries in the table give χ^2_{α} values, where α is the area or probability in the upper tail of the chi-squared distribution. For example, with ten degrees of freedom and a .01 area in the upper tail, $\chi^2_{0.1} = 23.209$

Degrees of freedom	Area in upper tail									
	.995	.99	.975	.95	.90	.10	.05	.025	.01	.005
1	.000	.000	.001	.004	.016	2.706	3.841	5.024	6.635	7.879
2	.010	.020	.051	.103	.211	4.605	5.991	7.378	9.210	10.597
3	.072	.115	.216	.352	.584	6.251	7.815	9.348	11.345	12.838
4	.207	.297	.484	.711	1.064	7.779	9.488	11.143	13.277	14.860
5	.412	.554	.831	1.145	1.610	9.236	11.070	12.832	15.086	16.750
6	.676	.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.647	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188

Critical value = 9.210

p-value $\approx$ 0.075

Kruskal-Wallis Test --- Example 2

Conclusion (a): Since $5.254 < 9.210$ we do NOT Reject H_0 and conclude that there is no significant difference in absenteeism rates from one level of employee to another, at the 1% significance level

OR

Conclusion (b): Since $p\text{-value} \approx 0.075 (> 0.01)$ we do NOT Reject H_0 and conclude that there is no significant difference in absenteeism rates from one level of employee to another, at the 1% significance level